University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Digital Design Fundamentals Lab	**Group #**
Course Code: CMPE-222L	
Lab Number: 5	
Lab Title: Testing a clocked Integrated Circuit	
CLOs to be covered: CLO 1	

Lab Goals

- **Task 1:** Building a clocking circuit using an astable multivibrator (CLO1).
- **Task 2:** Operating a 4-bit binary counter integrated circuit (CLO1).

Equipment Required

- **ICs:** 555 timer (1x), 7447 (1x), 7493 (1x), Seven segment display (1x)
- **Resistors:** 100Ω (7x) , 100kΩ (4x)
- **Capacitors:** 10µF (2x)
- **Tools:** IC puller (1x), Wire cutter(1x), Nose pliers (1x)
- **Other:** Jumper wires (as needed)

Lab Evaluation

Tasks	CLOs	Maximum points	Score (%)	Points scored
Task 1	1	3		
Task 2a	1	2		
Task 2b	1	4		
Task 2c	1	1		
Total points scored	**CLO1:**	**CLO2:**	**CLO3:**	

Rubrics for Lab Evaluation

Evaluation	Score (%)	Level	Rubric	CLOs
Completed	**100%**	L1	Completed the task correctly.	All
Great attempt	**75%**	L2	Almost completed the task, with minor deficiencies	2,3
Good attempt	**50%**	L3	Did not complete the task but managed to reasonably attempt it.	All
Basic attempt	**25%**	L4	Did not complete the task but managed to basically attempt it.	All
Did not attempt	**0%**	L5	Did not attempt the task.	All

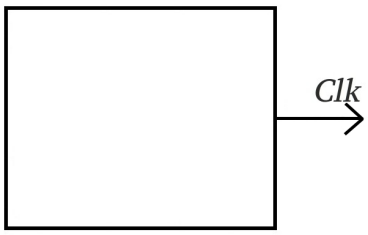
Building a clocking circuit using an astable multivibrator

Task 1 [CLO1, 3 Points]

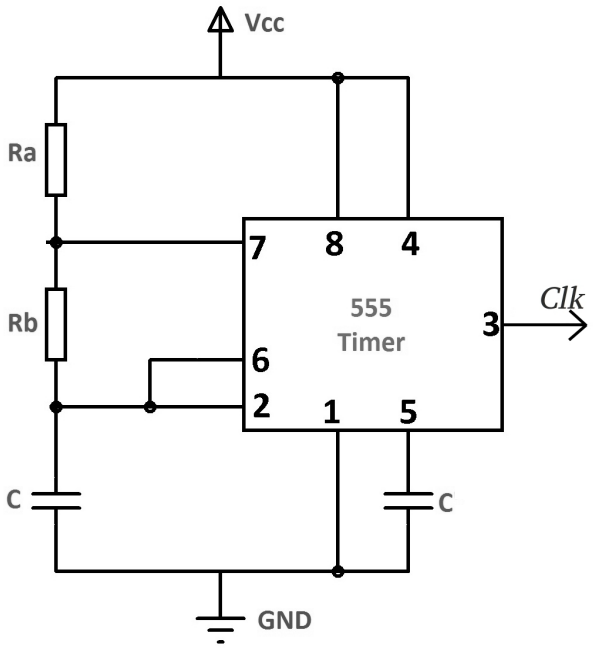
In this task you will be making a clocking circuit (binary stopwatch) with a time period of approximately 1 second. The circuit is implemented using a 555 timer IC (whose pinout is shown in the 'Pinout' section at the end), along with some resistors and capacitors. The figure below shows the prototype and implementation of the clocking circuit, technically called an **astable multivibrator**. (Note that Ra = Rb = 50kΩ, and C = 10μF, and that a 50kΩ resistor can be made by connecting two 100kΩ resistors in parallel). Wire up the circuit as shown in the **Implementation** below.

This circuit produces a binary clock signal (*Clk*) as an output, with a time period of approximately 1 second, (and a duty cycle of approximately 67 percent). LED 1 can be used to display this output. Connect the output *CLK* to LED 1, and show it to the lab instructor. (LED 1 should blink with a time period of approximately 1 second).

Prototype



Implementation



Operating a 4-bit binary counter integrated circuit

Task 2a [CLO1, 2 Points]

In this task you will operate the 7493 integrated circuit, which is a 4-bit binary counter. Insert this IC into the breadboard. The pinout of the 7493 IC is shown in the 'Pinout' section at the end. Connect the QA pin of this chip to CLKB. Connect the RST1 and RST2 pins to the ground rail (logic 0) of the trainer board. The 4-bit output of this counter comes from the pins labeled QD, QB, QC and QA respectively. Connect these four pins to **four consecutive LEDs**, so that you can see the binary output of the counter while it is counting. (It might be a good idea to use LEDs 1, 2, 3 and 4 for showing the (binary) outputs QD, QB, QC and QA respectively).

The counter should now start counting as soon as a clocking signal is applied to the CLKA pin. Connect the *CLK* output of the astable multivibrator **from Task 1** to CLKA. The count should cycle from zero to fifteen, (0000 to 1111 in binary), and be visible on the LEDs (1, 2, 3, and 4). Show this output to the lab instructor.

Task 2b [CLO1, 4 Points]

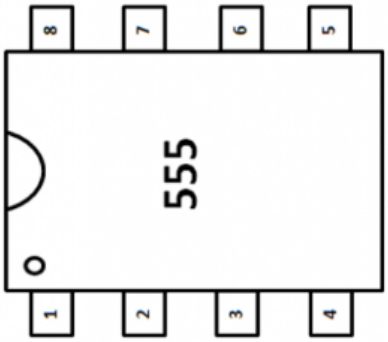
In this task you will be displaying the 4-bit output of the 7493 IC on a seven segment display unit. The pinout of the 7447 IC (seven segment display driver) and that of a seven segment display unit are shown in the 'Pinout' section at the end. **Interface the driver to the seven segment display unit using 100 Ohm resistors, like it was done in Lab1**. Now connect the the 4-bit output of the binary counter to the input of the driver. (The pins labeled QD, QB, QC, and QA on the 7493 IC have to be connected to the pins labeled D, C, B, and A on the 7447 IC respectively). The 4-bit output of the 7493 IC should now show up on the seven segment display unit. You should observe that when the output of the counter is from zero to nine (0000 to 1001 in binary), the seven segment display will display this output as a decimal digit. However, when the output of the counter goes from ten to fifteen (from 1010 to 1111 in binary), the seven segment display should display some garbage character. Show this to your lab instructor.

Task 2c [CLO1, 1 Points]

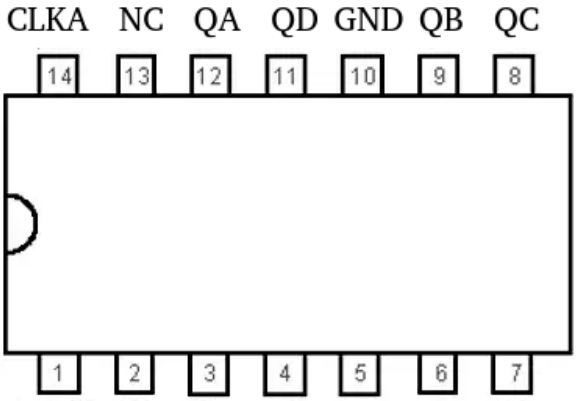
In this task you will limit the count of the 7493 IC to cycle between 0 and 9 (0000 and 1001 in binary). Connect QD to RST 1, and QB to RST2 on the IC. The chip should now count from 0 to 9 and go back to 0. This will be displayed on the seven segment display unit. Show this to the lab instructor.

Pinout

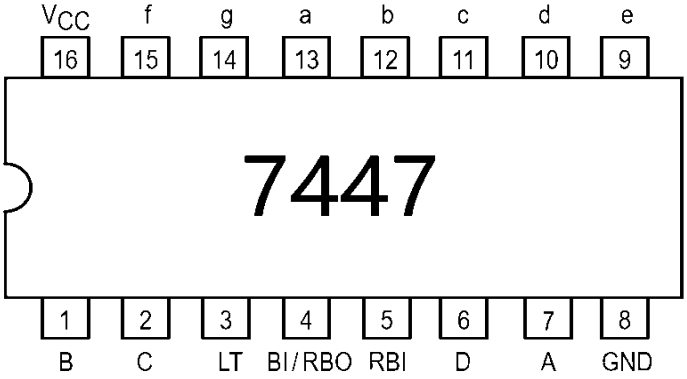
555 Timer IC



7493 IC



7447 IC



Seven-segment display

